

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO4: *Implement best security practices for IP, email, and web service using PGP, S/MIME for enhancing email Security.CO (BL3, BL6)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 25

Annexure A Comparative Analysis

Code of Conduct:

I Kanmani L/192511414 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A
Comparative Analysis

1. Apply **SSL/TLS and IPsec protocols** to a secure communication scenario and determine which provides better security for web-based and network-layer protection.
2. Design a secure email system by comparing **end-to-end encryption and gateway-based encryption**, and justify the most suitable approach for enterprise use.
3. Apply **HMAC and Digital Signature techniques** to a data transmission scenario and analyze which method provides better authentication and integrity.
4. Create a **spam detection system** by comparing supervised learning and unsupervised learning approaches, and evaluate their effectiveness in real-world email filtering.
5. Design a secure communication framework by comparing **AES and RSA encryption techniques**, and determine their roles in achieving confidentiality and efficiency.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

1. SSL/TLS and IPsec are security protocols used to protect data during network communication. SSL/TLS mainly secures application-layer communication such as web browsing, while IPsec protects IP packets at the network layer.

Consider an organization where employees access an online banking/web application and also connect to the company's internal network remotely.

- SSL/TLS: Used between the employee's browser and the web server.
- IPsec: Used to establish a secure VPN connection between the employee and the organization's network.

SSL/TLS Operation

1. Client connects to the web server.
2. Server provides its digital certificate.
3. Client verifies the certificate.
4. A secure session key is established.
5. Data exchanged between browser and server is encrypted.

Protection provided: Confidentiality, authentication, and integrity of web/application data.

IPsec Operation

1. A secure connection is established between the client and network gateway.
2. IPsec authenticates the communicating endpoints.
3. IP packets are encrypted and protected.
4. The packets travel securely through the untrusted network.
5. The receiving gateway decrypts and forwards the original packets.

Protection provided: Confidentiality, authentication, and integrity at the network layer.

Comparison

Feature	SSL/TLS	IPsec
Layer	Transport/Application interface	Network layer
Main use	Web applications, HTTPS	VPNs, network-to-network security
Protection	Application data	Entire IP packets
Authentication	Digital certificates	Certificates/pre-shared keys
Best suited for	Web-based communication	Network/VPN protection

Conclusion

For web-based protection, SSL/TLS is more appropriate because it directly secures HTTPS communication between the browser and web server. For network-layer protection, IPsec is better because it can secure entire IP packets and protect VPN or site-to-site communication.

Therefore, neither is universally "more secure"; SSL/TLS is the better choice for web applications, while IPsec is the better choice for securing network-layer communication.

2. Secure Email System: End-to-End vs Gateway-Based Encryption

Introduction

Email encryption protects sensitive information from unauthorized access. End-to-end encryption (E2EE) protects the message from the sender to the recipient, while gateway-based encryption encrypts and decrypts messages through an organization's security gateway.

Comparison

Feature	End-to-End Encryption	Gateway-Based Encryption
Encryption	Sender to recipient	Sender/gateway to gateway/recipient
Key management	Managed by users/devices	Centrally managed
Security	Very high	High
Enterprise control	Lower	Higher
Administration	More complex	Easier
Monitoring/scanning	Limited	Easier
Best use	Highly confidential communication	Large enterprises

Recommendation

For most enterprises, gateway-based encryption is more suitable because centralized management makes certificate/key management, policy enforcement, malware scanning, and compliance easier. However, E2EE should be used for highly sensitive communications where maximum message confidentiality is required.

3. HMAC vs Digital Signature

Introduction

HMAC (Hash-based Message Authentication Code) and digital signatures provide message authentication and integrity. HMAC uses a shared secret key, while digital signatures use a private/public key pair.

Scenario

Suppose a bank sends transaction data to another server.

Using HMAC:

1. Sender and receiver share a secret key.
2. Sender generates an HMAC for the transaction.
3. Receiver calculates the HMAC using the same key.
4. Matching HMACs confirm integrity and authenticity.

Using Digital Signature:

1. Sender hashes the transaction.
2. Sender signs the hash using a private key.
3. Receiver uses the sender's public key to verify the signature.
4. Successful verification confirms authenticity and integrity.

Comparison

Feature	HMAC	Digital Signature
Key type	Shared secret	Private/public key
Integrity	Yes	Yes
Authentication	Yes	Yes
Non-repudiation	No	Yes
Speed	Faster	Slower
Key distribution	More difficult at large scale	Easier

Conclusion

HMAC is faster and efficient when both parties can securely share a secret key. Digital signatures provide stronger authentication and non-repudiation, making them more suitable when communication involves multiple parties or legal verification. Therefore, digital signatures provide broader authentication capabilities, while HMAC is better for high-speed trusted systems.

4. Spam Detection: Supervised vs Unsupervised Learning

Introduction

Spam detection uses machine-learning techniques to classify emails as spam or legitimate. Supervised learning uses labeled email data, while unsupervised learning discovers patterns without predefined labels.

Supervised Learning

1. Collect labeled spam and legitimate emails.
2. Extract features such as keywords, sender information, URLs, and email frequency.
3. Train a classifier such as Naive Bayes or Logistic Regression.
4. Test the model on new emails.
5. Classify emails as spam or legitimate.

Advantages: High accuracy when sufficient quality training data is available.

Unsupervised Learning

1. Collect emails without labels.
2. Extract relevant email features.
3. Group similar emails using clustering algorithms such as K-Means.
4. Identify clusters containing suspicious or spam-like patterns.
5. Classify new emails based on discovered patterns.

Advantages: Does not require manually labeled training data and can discover new patterns.

Comparison

Feature	Supervised Learning	Unsupervised Learning
Labeled data	Required	Not required
Accuracy	Generally higher	Generally lower
New spam detection	Depends on training data	Can discover new patterns
Training	More expensive	Less dependent on labeling
Common methods	Naive Bayes, SVM	K-Means, clustering

Conclusion

Supervised learning is generally more effective for real-world spam filtering because it can achieve high accuracy using well-labeled datasets. However, combining supervised and unsupervised techniques can improve detection of new and previously unseen spam patterns.

5. AES vs RSA for Secure Communication

Introduction

AES and RSA are cryptographic algorithms with different purposes. AES is a fast symmetric encryption algorithm, while RSA is an asymmetric algorithm used mainly for secure key exchange and digital signatures.

Secure Communication Framework

1. Generate a random AES session key.
2. Encrypt the actual message/data using AES.
3. Encrypt the AES session key using the receiver's RSA public key.
4. Send the encrypted data and encrypted session key.
5. Receiver uses the RSA private key to recover the AES key.
6. Receiver uses the AES key to decrypt the data.

Comparison

Feature	AES	RSA
Type	Symmetric	Asymmetric
Keys	One shared key	Public + private key
Speed	Very fast	Relatively slow
Large data encryption	Suitable	Not efficient
Key exchange	Not by itself	Suitable
Digital signatures	No	Yes

Roles in Secure Communication

AES: Provides efficient confidentiality for large amounts of data such as files, emails, and web traffic.

RSA: Provides secure key exchange and can provide authentication through digital signatures.

Conclusion

AES and RSA are most effective when used together. AES handles bulk data encryption efficiently, while RSA securely protects the AES session key and supports authentication. This hybrid approach provides both strong confidentiality and high performance, and is widely used in secure communication systems.